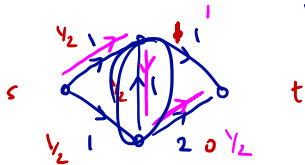


Primal Dual and Augmenting Paths

Improves the dual by finding an augmenting path: an undirected path restricted to use edge in forward direction only if that edge is unsaturated ($f_{ij} < b_{ij}$), and using an edge in bw direction, only if that edge had +ve flow ($f_{ij} > 0$).



$$f \leq b$$

$$f \geq 0$$

$$f_{\text{new}} = f + \theta \bar{f}$$

Key: What happens when

we cannot find an aug. path? $\rightarrow$ Optimal.

$\hookrightarrow$ From LP Duality: Max Flow / Min Cut.

Cuts and Min-Cuts

Def'n: s - t cut is a subset

$$W \subseteq V, \text{ s.t. } s \in W, t \in \bar{W}$$

Note: a cut is also sometimes defined

as edges b/w W & $\bar{W}$

$\delta^-(v)$ = edges/arcs terminating in v

$\delta^-(W)$ = edges from W to $\bar{W}$

Capacity of cut: $c(W, \bar{W}) = \sum_{\substack{(i,j): \\ i \in W, j \in \bar{W}}} b_{ij}$

Max Flow LP & Dual

max: v

st: $Af + dv = 0$ $\downarrow$ *

$$f \leq b$$

$$f \geq 0$$

$$d_i = \begin{cases} -1 & i=s \\ +1 & i=t \\ 0 & \end{cases}$$

$$\min: \sum \gamma_{ij} b_{ij} + 0$$

st: $p_i - p_j + \gamma_{ij} \geq 0$

$$-p_s + p_t \geq 1$$

$$\gamma_{ij} \geq 0$$

$$p \text{ free.}$$

Given an s-t cut: $(W, \bar{W})$ we can

define:

$$\gamma_{ij} = \begin{cases} 1 & \text{if } i \in W, j \in \bar{W} \\ 0 & \end{cases}$$

$$p_i = \begin{cases} 0 & i \in W \\ 1 & i \in \bar{W} \end{cases}$$

Cuts as Dual Solutions

$$\min: \sum \gamma_{ij} b_{ij} + 0$$

$$\text{st: } p_i - p_j + \gamma_{ij} \geq 0 \quad //$$

$$-p_s + p_t \geq 1$$

$$\gamma_{ij} \geq 0$$

p free.

Easy to check dual feasibility.

$$\text{key pt: } p_i - p_j + \gamma_{ij} \geq 0$$

$$\underline{\text{strict}} \quad \text{iff } i \in \bar{W}, j \in W$$

$$1 - 0 + 0 > 0$$

$$\gamma_{ij} = \begin{cases} 1 & \text{if } i \in W, j \in \bar{W} \\ 0 & \end{cases}$$

$$p_i = \begin{cases} 0 & i \in W \\ 1 & i \in \bar{W} \end{cases}$$

Max Flow = Min Cut

Thm: Value of s-t flow is

upper bounded by $C(W, \bar{W}) \quad \forall \text{ s-t cut.}$

Moreover, max flow = min cut and flow f

and cut $(W, \bar{W})$ are jointly optimal

$$f_{ij} = 0 \quad \forall (i,j) \in E \text{ st. } i \in \bar{W}, j \in W$$

$$f_{ij} = b_{ij} \quad \forall (i,j) \in E \text{ st. } i \in W, j \in \bar{W}$$

// C.S.